ChryslerScanner V240-USB-C-OBD-SD-A

259 components | 154 nets | 23 native KiCad pages | integrated SD revision

CCD interface

25 parts / page 2

File: 01_ccd.kicad_sch

PCI / J1850

12 parts / page 3

File: 02_pci.kicad_sch

SCI TX14 / TX9

9 parts / page 4

File: 03_sci-14-9.kicad_sch

SCI TX6 / TX12 + RX bias

12 parts / page 5

File: 04_sci-6-12.kicad_sch

SCI RX7 / RX15

9 parts / page 6

File: 05_sci-rx.kicad_sch

OBD line protection

11 parts / page 7

File: 06_obd-protection.kicad_sch

Vehicle power input

4 parts / page 8

File: 07_vehicle-input.kicad_sch

5 V / 3.3 V supplies

17 parts / page 9

File: 08_power-5v.kicad_sch

VPP boost supply

10 parts / page 10

File: 09_power-vpp.kicad_sch

VBB supply

10 parts / page 11

File: 10_power-vbb.kicad_sch

7.5 V supply

15 parts / page 12

File: 11_power-7v5.kicad_sch

SCI TX6 VBB / VPP drivers

9 parts / page 13

File: 12_program-6.kicad_sch

SCI TX9 VBB / VPP drivers

9 parts / page 14

File: 13_program-9.kicad_sch

SCI TX12 VBB / VPP drivers

9 parts / page 15

File: 14_program-12.kicad_sch

SCI TX14 VBB / VPP drivers

9 parts / page 16

File: 15_program-14.kicad_sch

ESP32 + RF

16 parts / page 17

File: 16_mcu.kicad_sch

Voltage monitoring

12 parts / page 18

File: 17_adc.kicad_sch

I/O expander + I2C

7 parts / page 19

File: 18_control.kicad_sch

USB + reset / boot

19 parts / page 20

File: 19_usb.kicad_sch

Status LED drivers

16 parts / page 21

File: 20_leds.kicad_sch

OBD connector

1 parts / page 22

File: 21_obd.kicad_sch

Integrated SD logger

18 added parts / page 23

File: 22_sd-logger.kicad_sch

Open a sheet to inspect symbols and wires. Matching signal names connect across sheets.

Preserves USB-C, ATTEND OBD-II and scanner functions. SD hardware added; firmware remains deferred.

Electrical review: U3 / U4 / U8 pin 5 requires GND; currently NC in the source.

Passive pin types: connectivity checked, independent electrical review still required.

ChryslerScanner / atopile

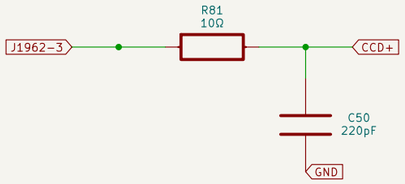
Sheet: /
File: ChryslerScanner_V240_SD.kicad_sch

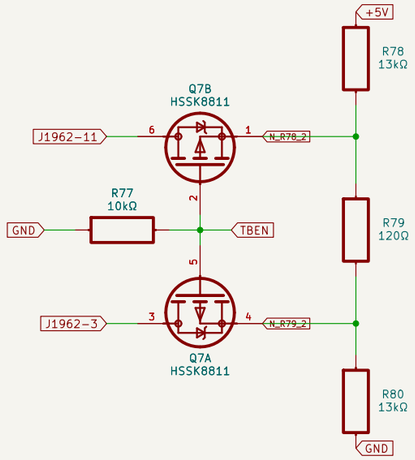
Title: ChryslerScanner V240-USB-C-OBD | circuit overview

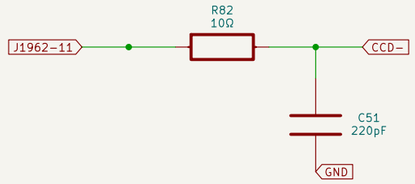
Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 1/23

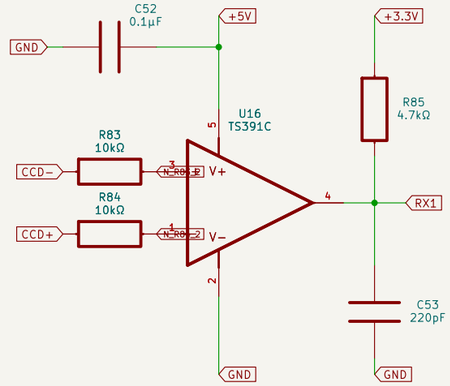
CCD interface

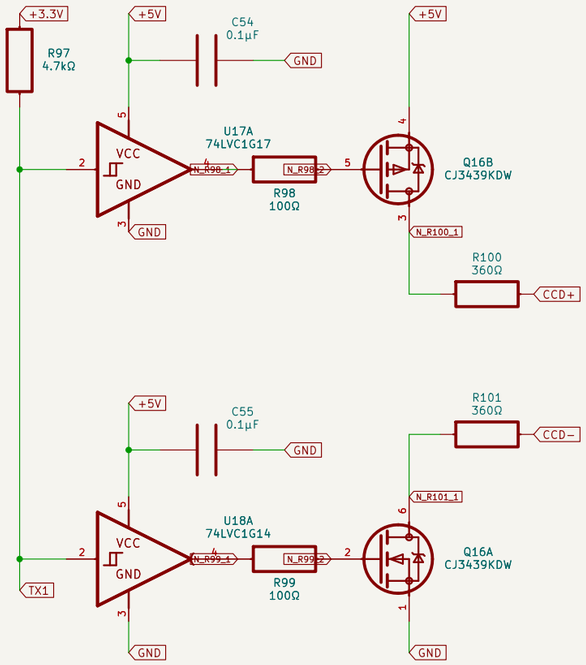
Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

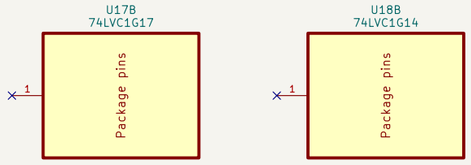












Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

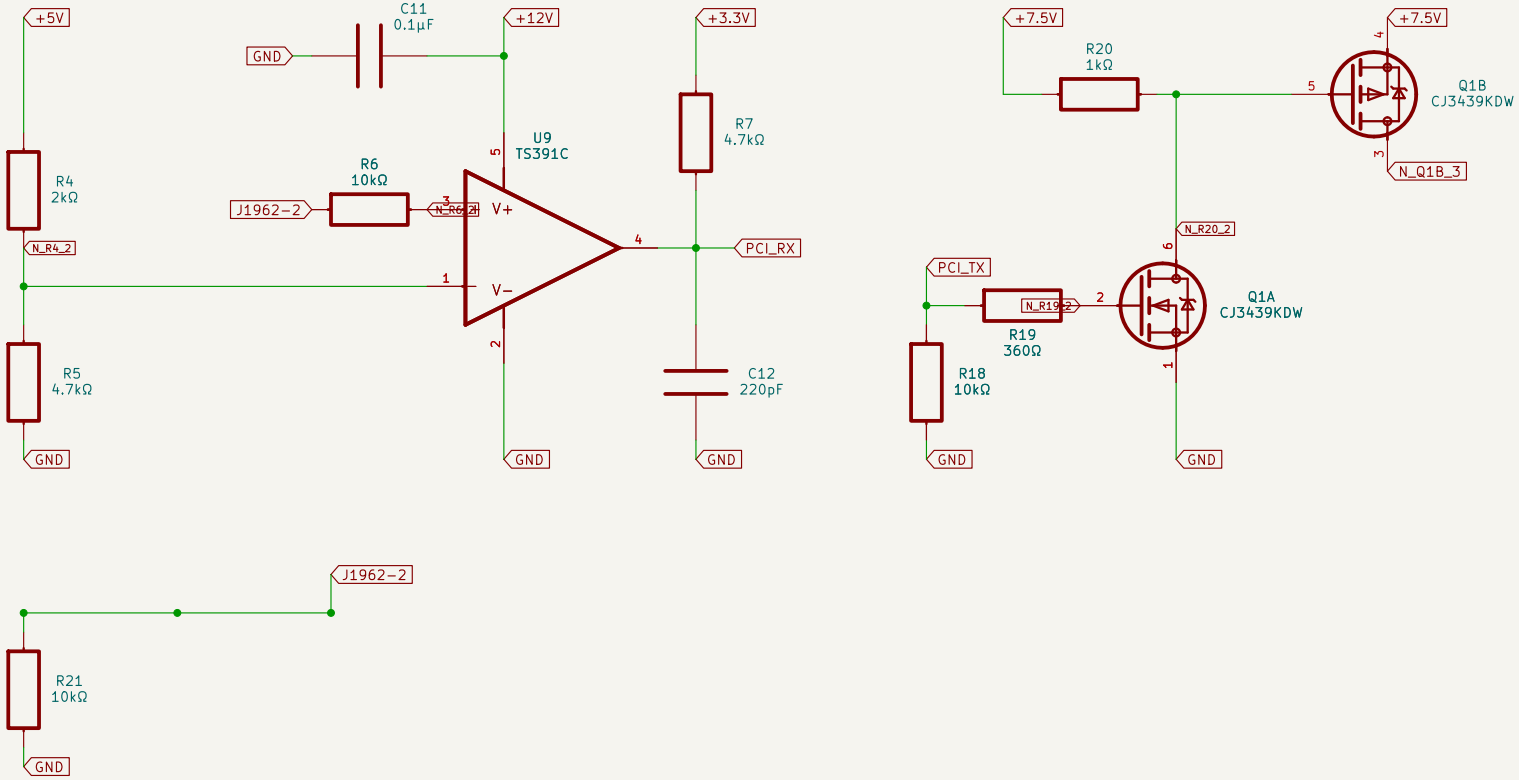
Sheet: /CCD interface/
File: 01_ccd.kicad_sch

Title: CCD interface

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 2/23

PCI / J1850

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

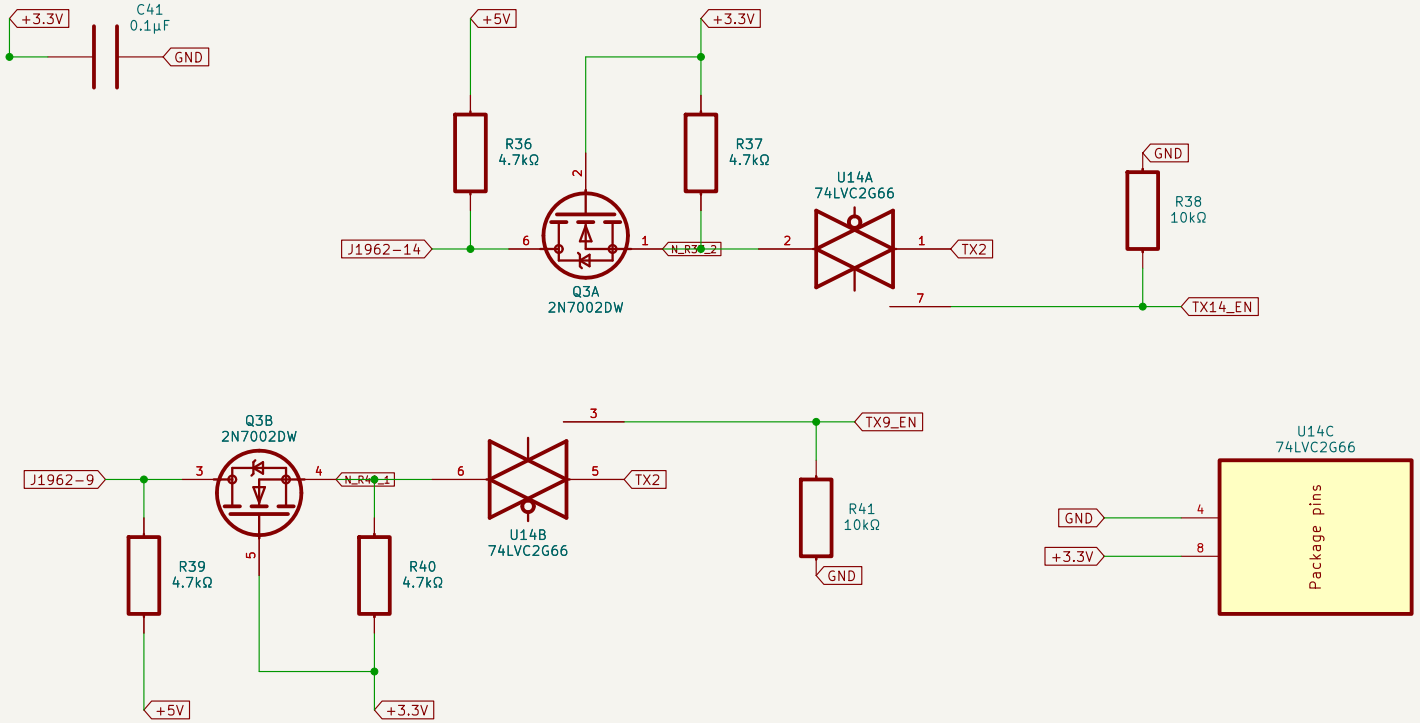


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /PCI / J1850/		
File: 02_pci.kicad_sch		
Title: PCI / J1850		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 3/23

SCI TX14 / TX9

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



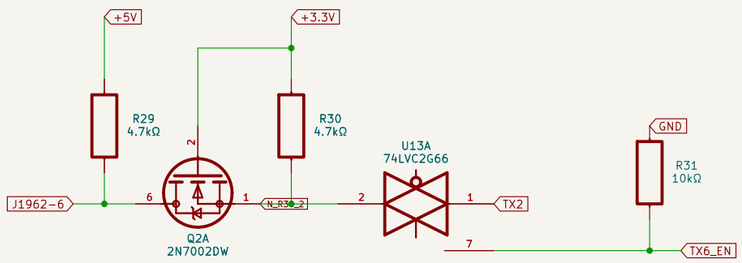
Atopile source / editable KiCad view / passive pins require electrical review.

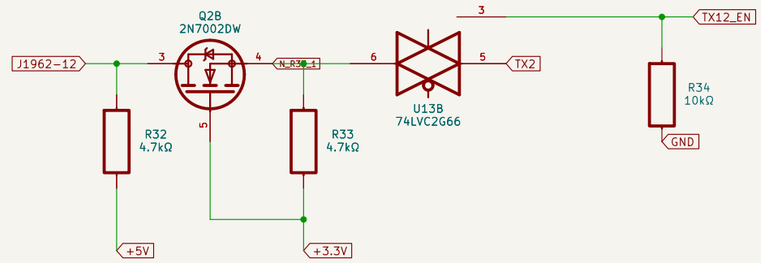
ChryslerScanner / atopile		
Sheet: /SCI TX14 / TX9/		
File: 03_sci-14-9.kicad_sch		
Title: SCI TX14 / TX9		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 4/23

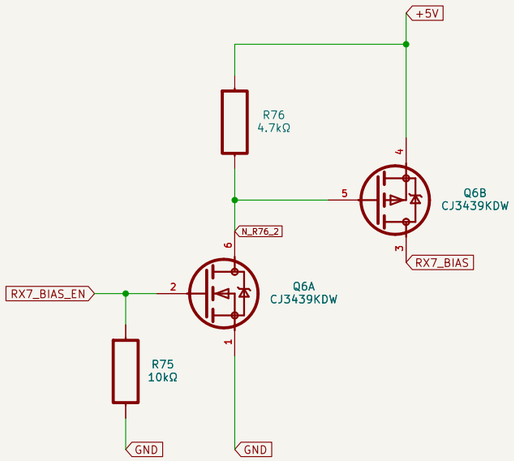
SCI TX6 / TX12 + RX bias

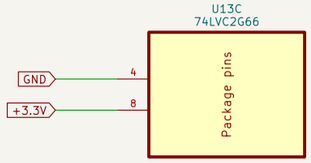
Matching signal names connect between sheets. Values and package pins follow the atopile netlist.











Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

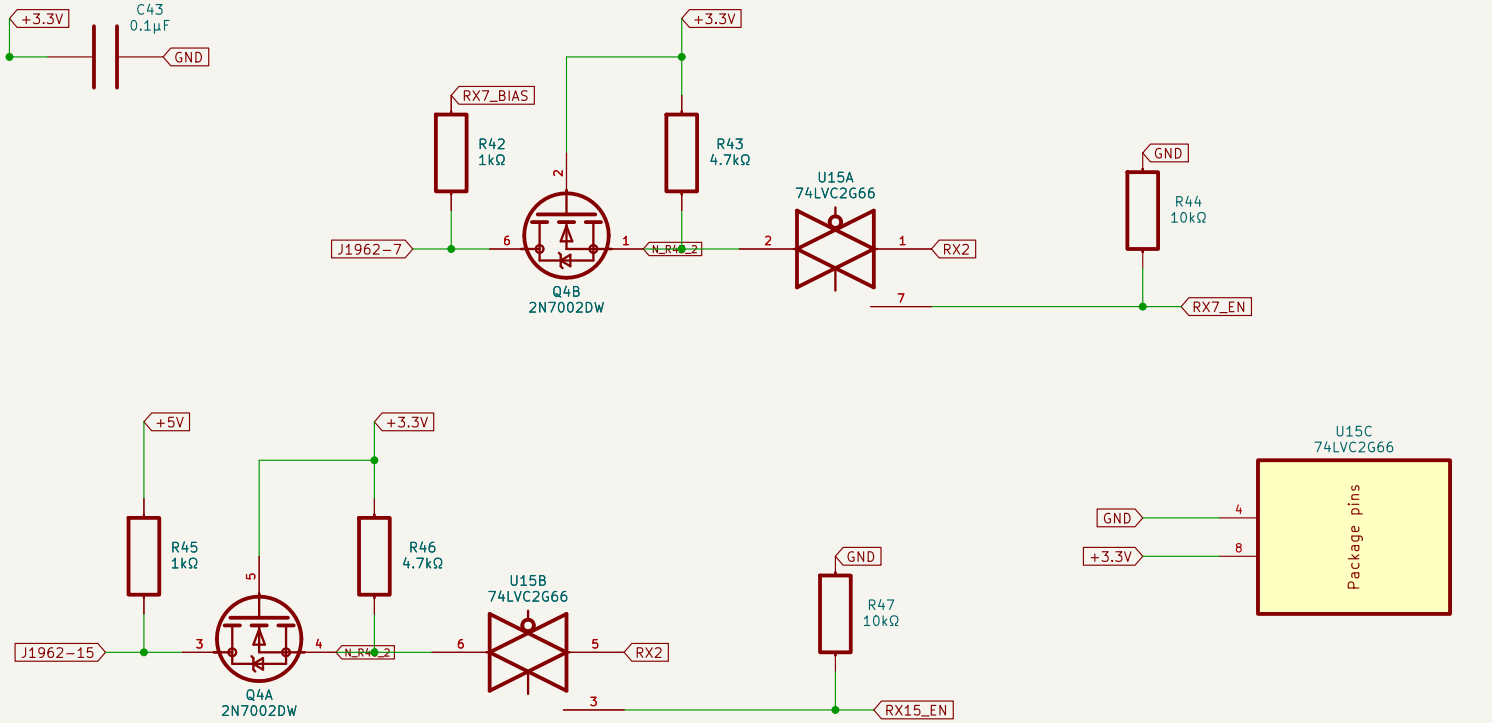
Sheet: /SCI TX6 / TX12 + RX bias/
File: 04_sci-6-12.kicad_sch

Title: SCI TX6 / TX12 + RX bias

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 5/23

SCI RX7 / RX15

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

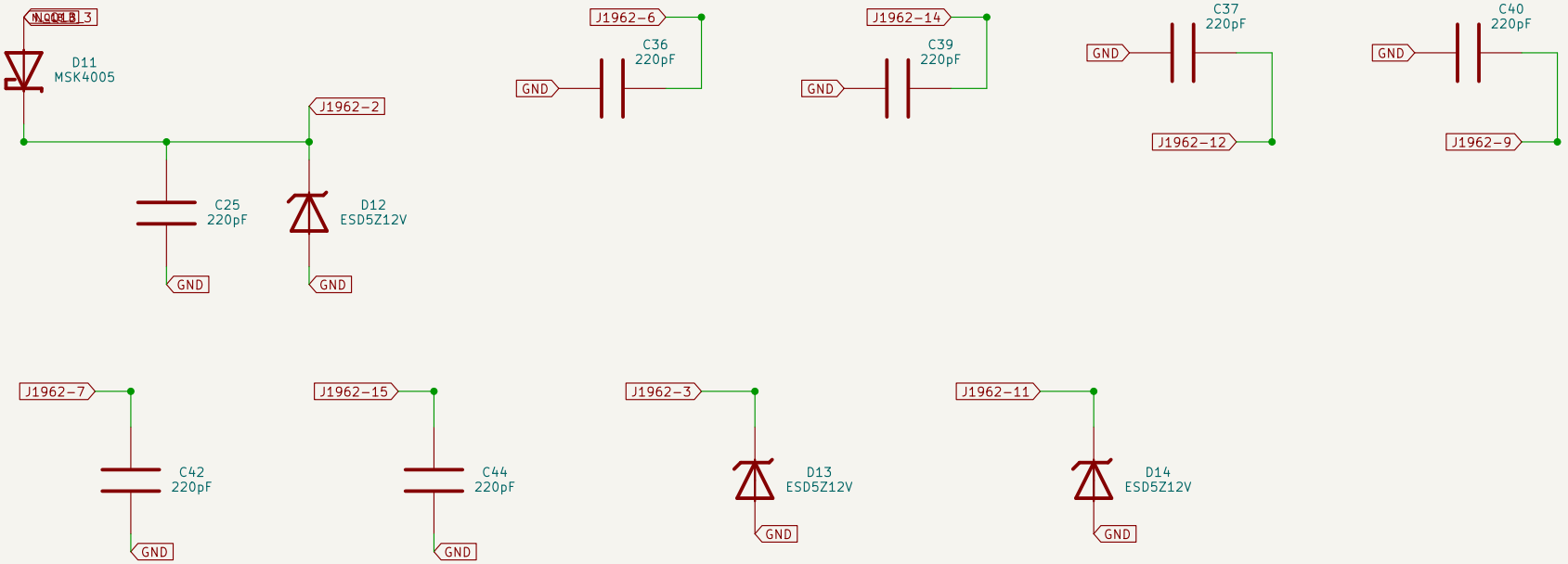


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /SCI RX7 / RX15/		
File: 05_sci-rx.kicad_sch		
Title: SCI RX7 / RX15		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 6/23

OBD line protection

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

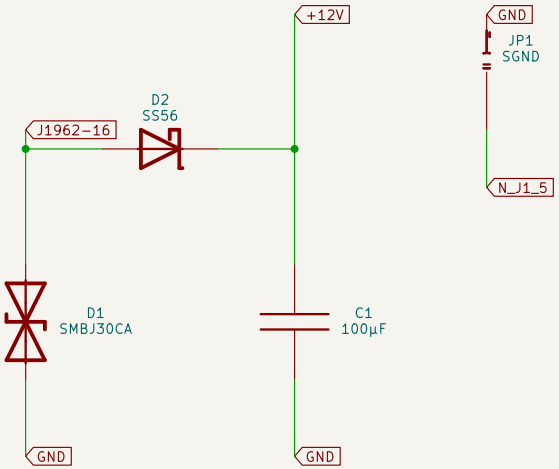
Sheet: /OBD line protection/
File: 06_obd-protection.kicad_sch

Title: OBD line protection

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 7/23

Vehicle power input

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

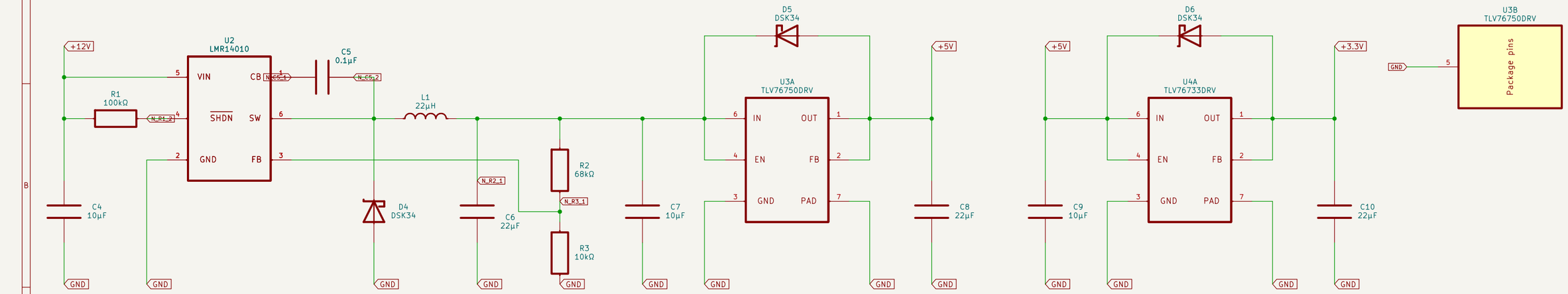
Sheet: /Vehicle power input/
File: 07_vehicle-input.kicad_sch

Title: Vehicle power input

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 8/23

5 V / 3.3 V supplies

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



REVIEW: U3 / U4 pin 5 is NC in the source; TLV767 DRV requires GND.

Source connectivity is preserved here. Correct this in atopile before routing.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

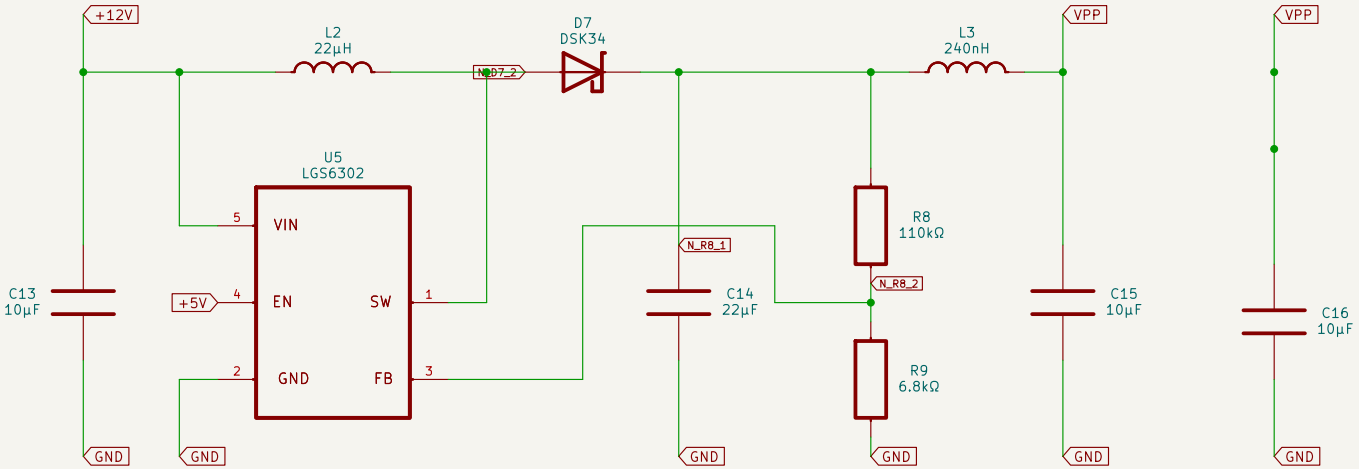
Sheet: /5 V / 3.3 V supplies/
File: 08_power-5v.kicad_sch

Title: 5 V / 3.3 V supplies

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 9/23

VPP boost supply

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

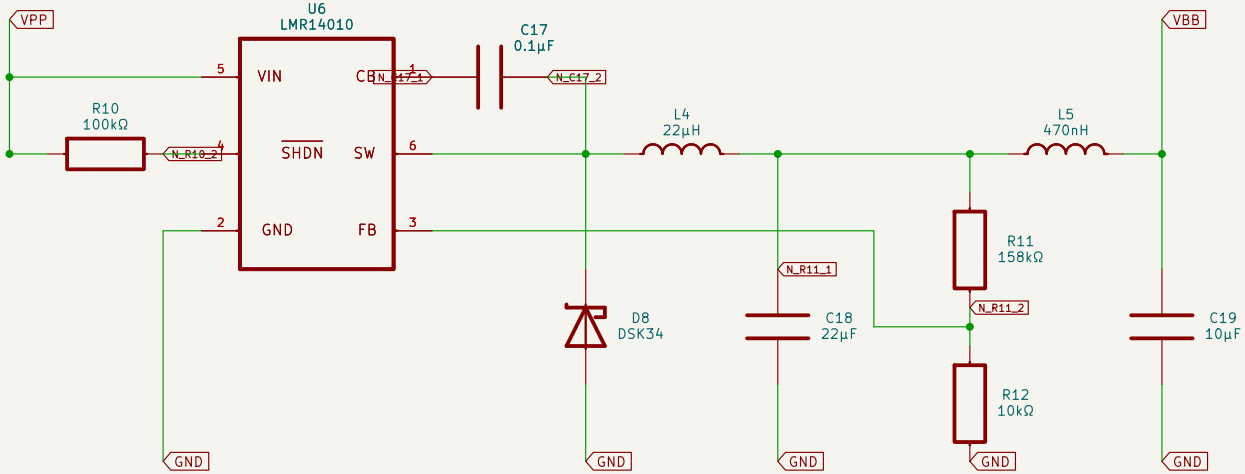
Sheet: /VPP boost supply/
File: 09_power-vpp.kicad_sch

Title: VPP boost supply

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 10/23

VBB supply

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

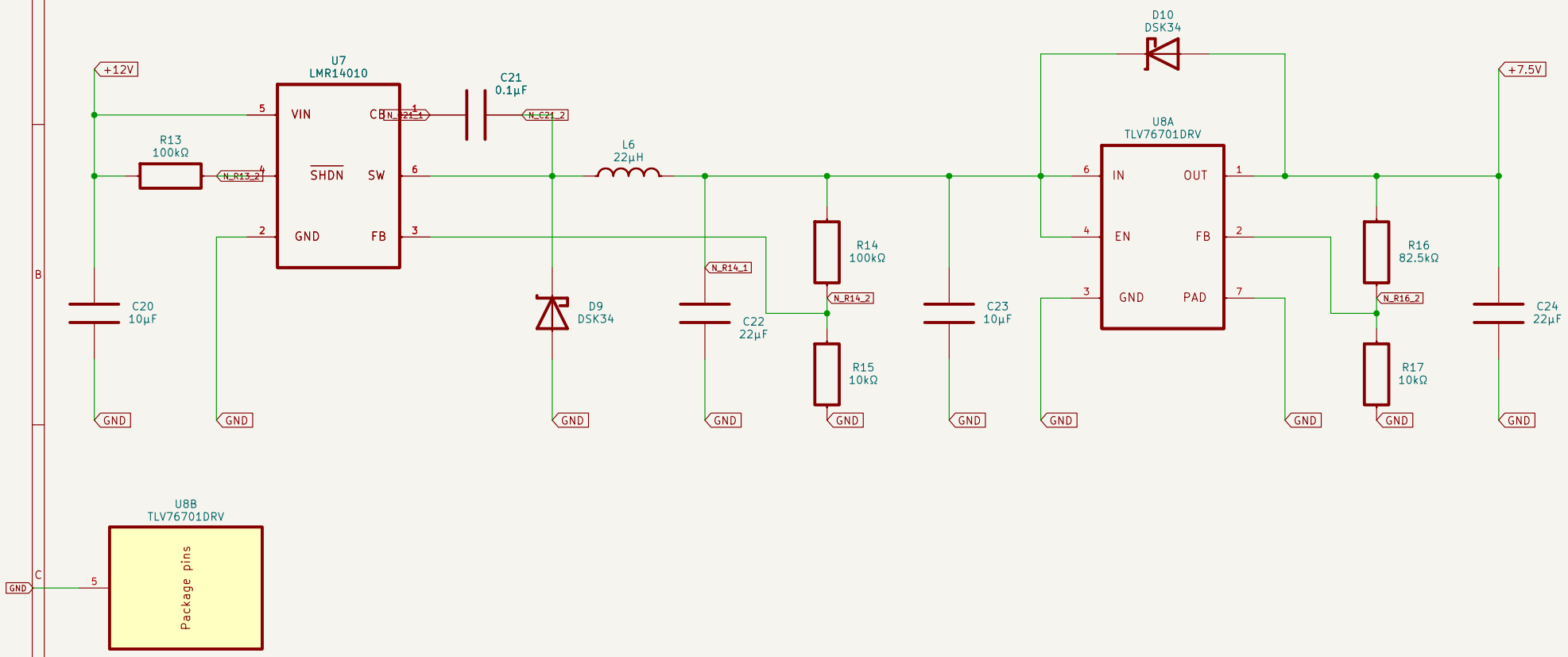


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /VBB supply/ File: 10_power-vbb.kicad_sch		
Title: VBB supply		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 11/23

7.5 V supply

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



REVIEW: U8 pin 5 is NC in the source; TLV767 DRV requires GND.

Source connectivity is preserved here. Correct this in atopile before routing.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

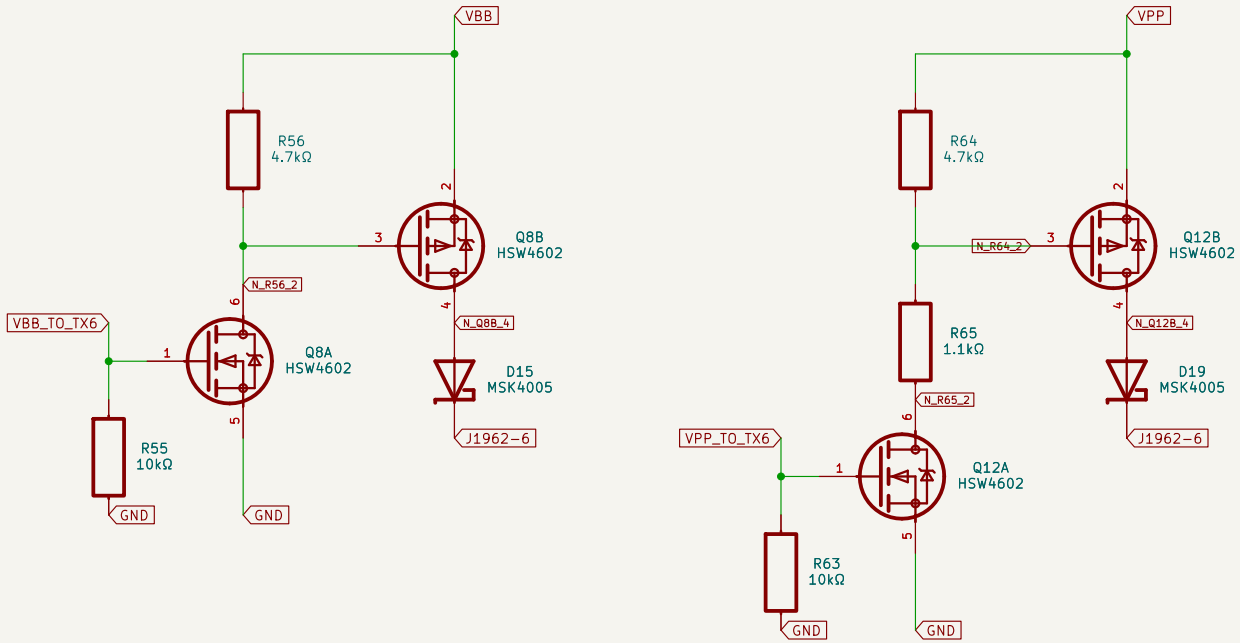
Sheet: /7.5 V supply/
File: 11_power-7v5.kicad_sch

Title: 7.5 V supply

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 12/23

SCI TX6 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

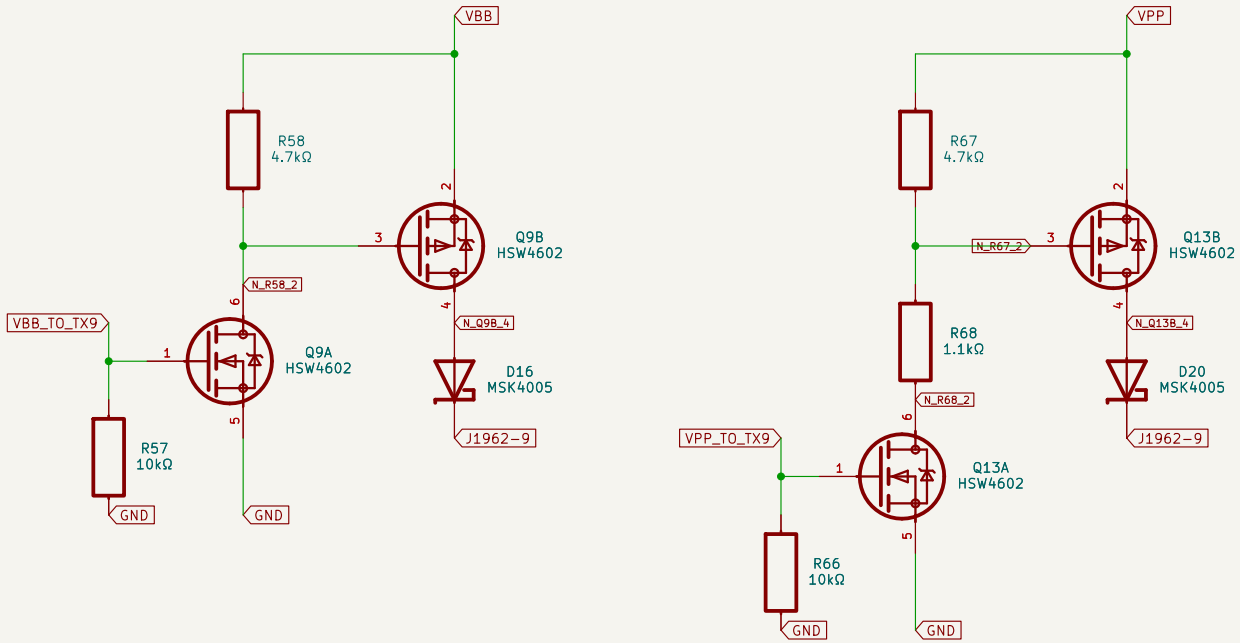


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /SCI TX6 VBB / VPP drivers/		
File: 12_program-6.kicad_sch		
Title: SCI TX6 VBB / VPP drivers		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 13/23

SCI TX9 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

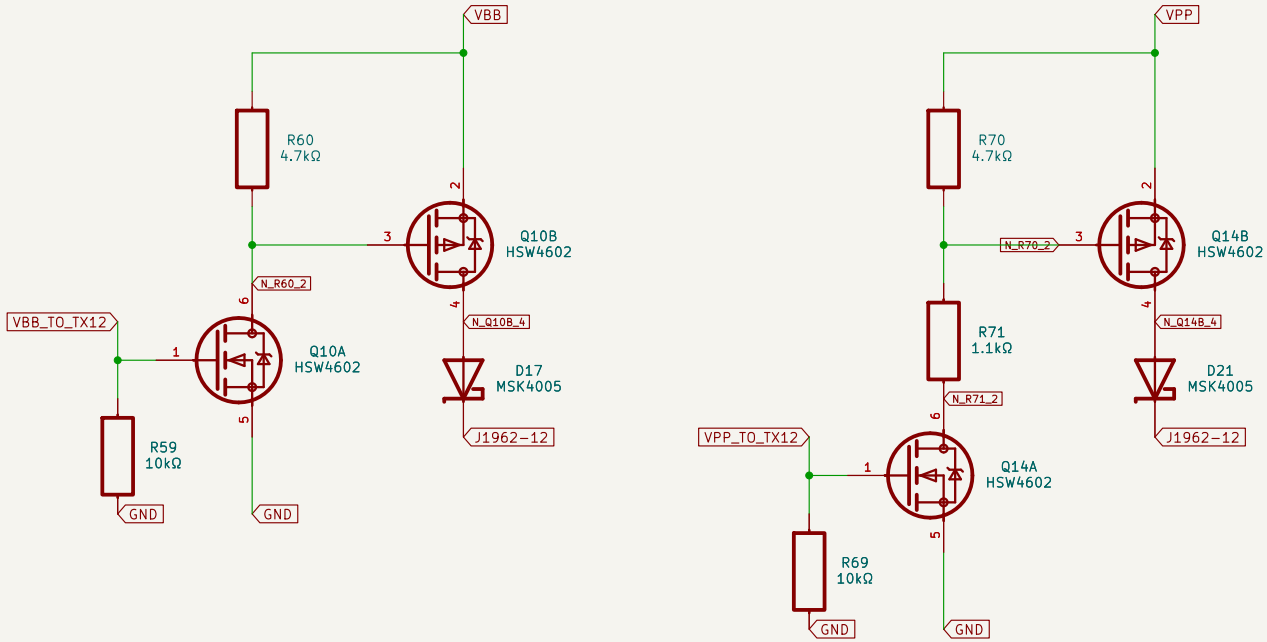
Sheet: /SCI TX9 VBB / VPP drivers/
File: 13_program-9.kicad_sch

Title: SCI TX9 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 14/23

SCI TX12 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

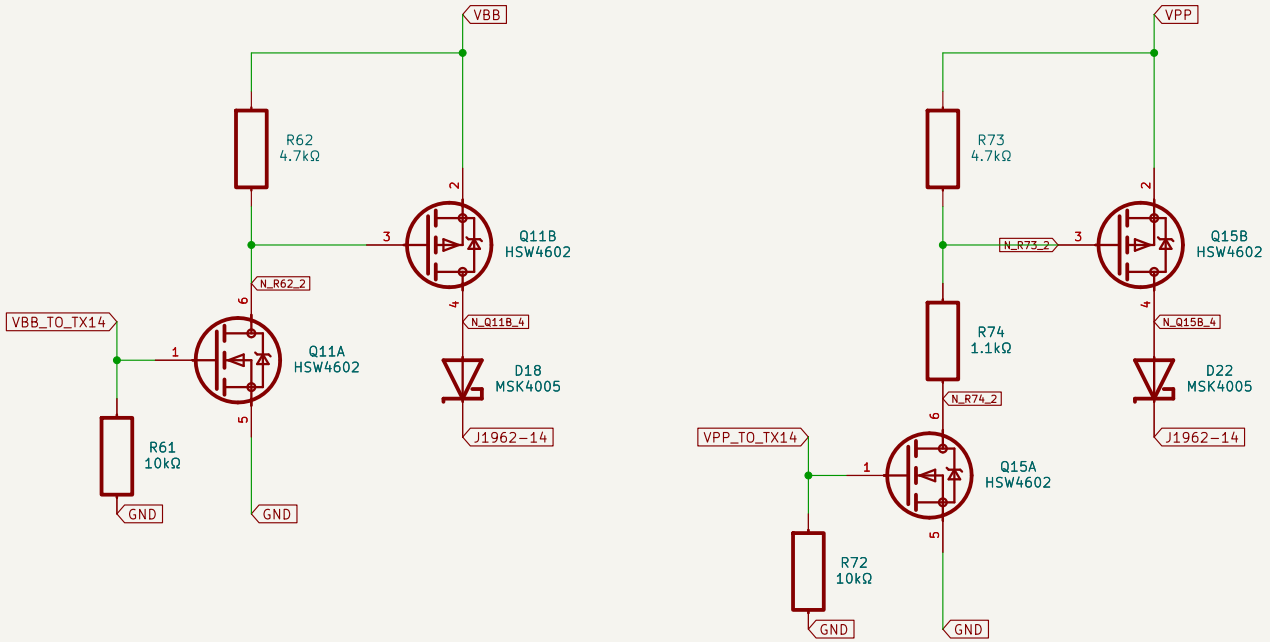
Sheet: /SCI TX12 VBB / VPP drivers/
File: 14_program-12.kicad_sch

Title: SCI TX12 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 15/23

SCI TX14 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

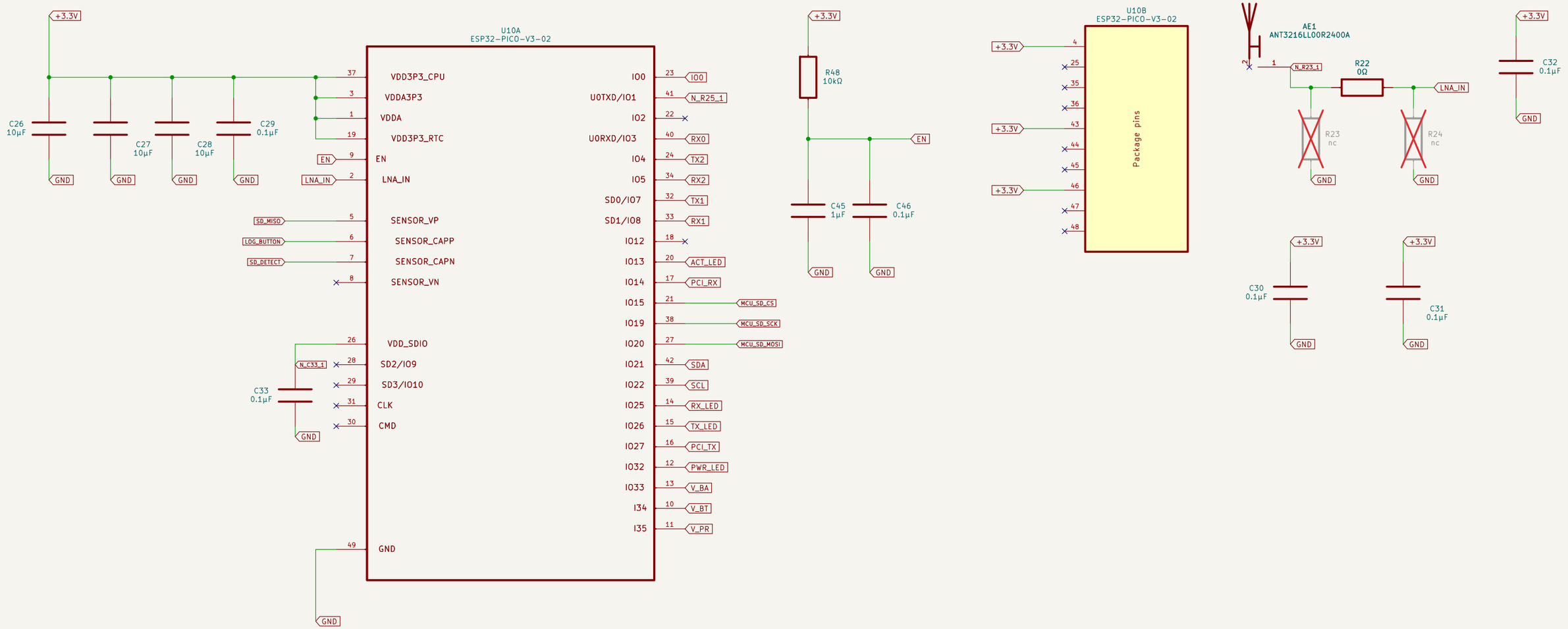
Sheet: /SCI TX14 VBB / VPP drivers/
File: 15_program-14.kicad_sch

Title: SCI TX14 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 16/23

ESP32 + RF

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



SD revision: GPIO19/20 SPI, GPIO36 MISO, GPIO15 CS, GPIO37 button, GPIO38 detect. Firmware deferred.

Assembly option: R23 and R24 are DNP.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

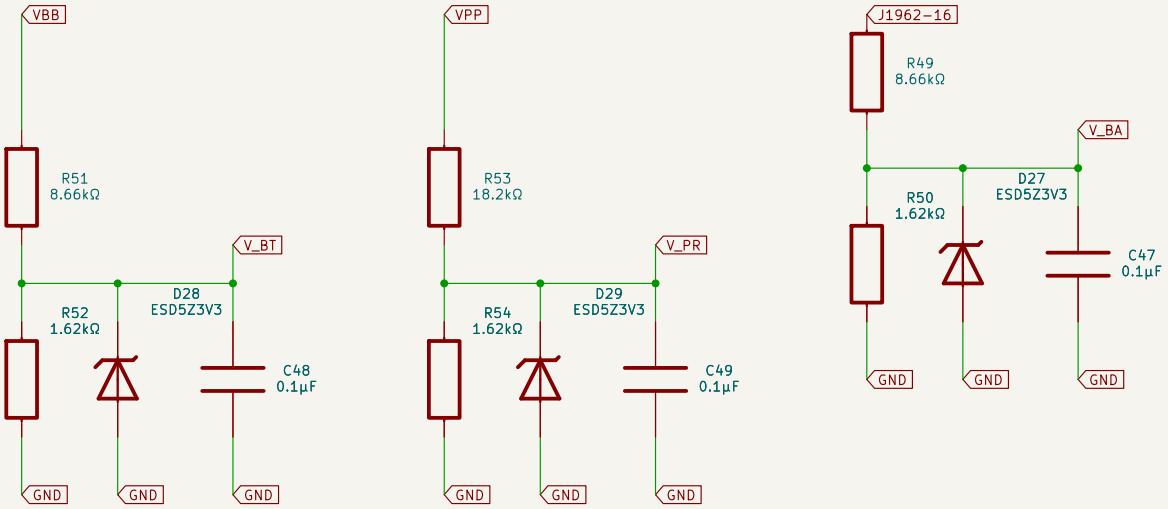
Sheet: /ESP32 + RF/
File: 16_mcu.kicad_sch

Title: ESP32 + RF

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 17/23

Voltage monitoring

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

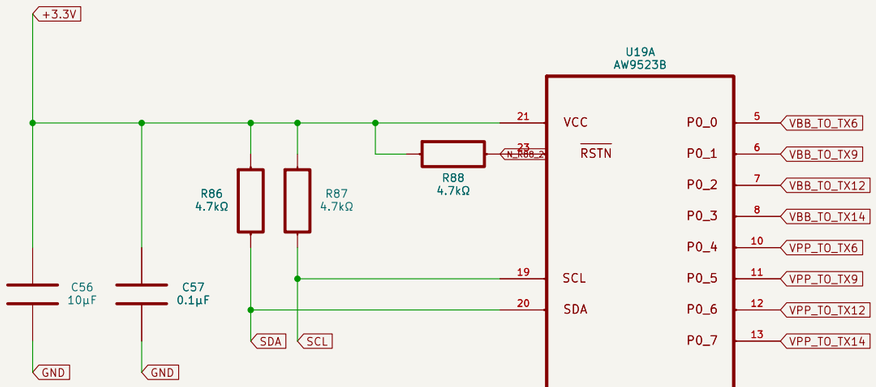
Sheet: /Voltage monitoring/
File: 17_adc.kicad_sch

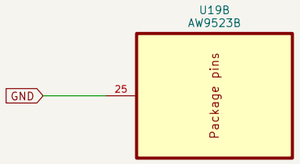
Title: Voltage monitoring

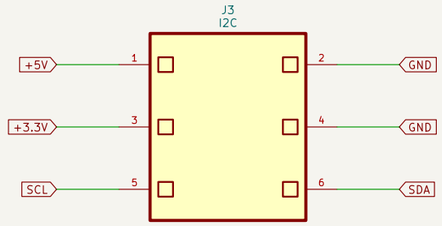
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 18/23

I/O expander + I2C

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.







Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

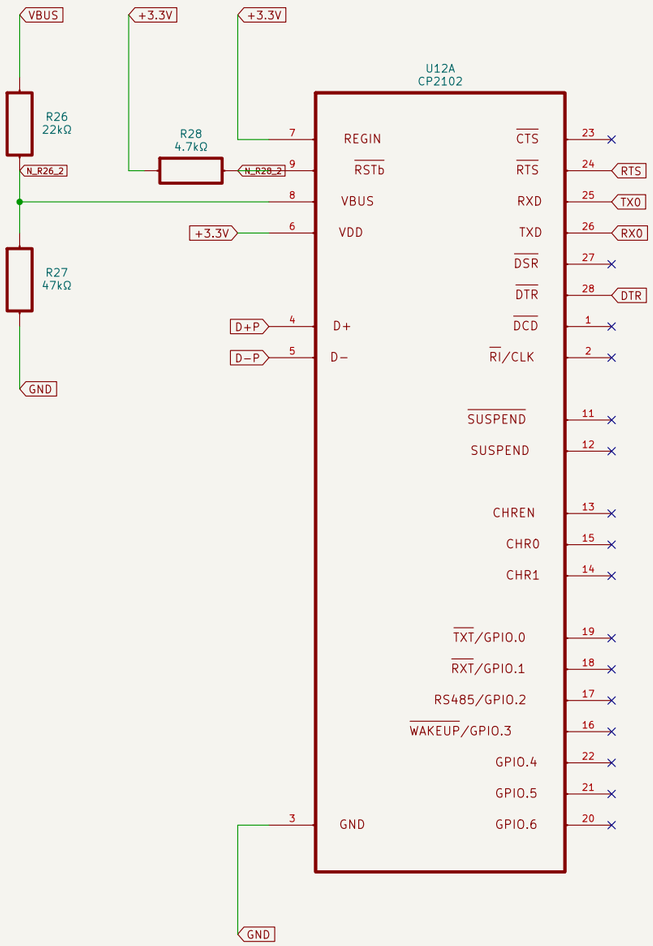
Sheet: /I/O expander + I2C/
File: 18_control.kicad_sch

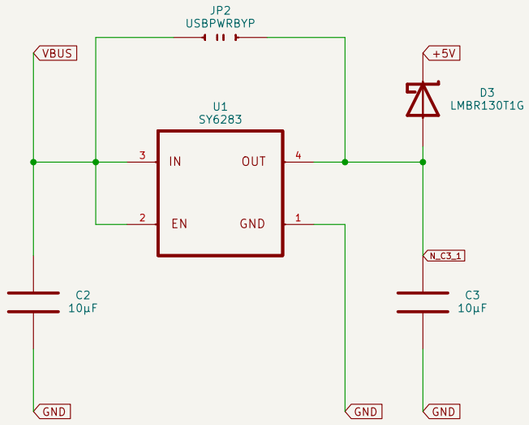
Title: I/O expander + I2C

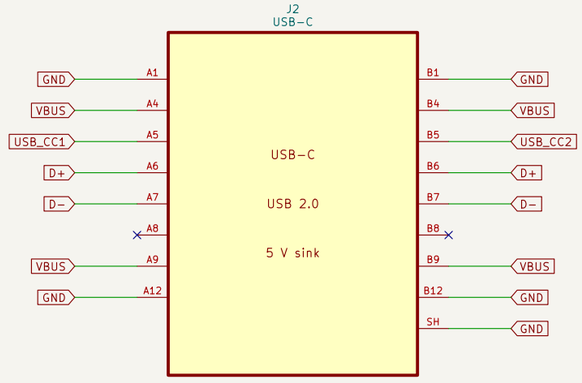
Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 19/23

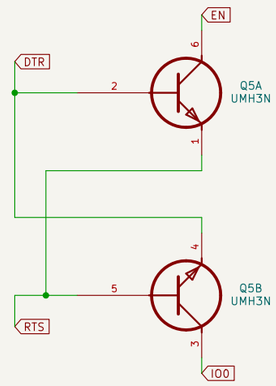
USB + reset / boot

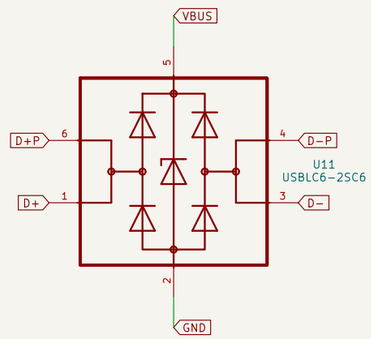
Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

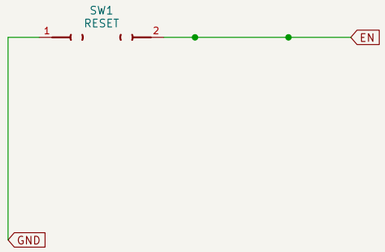


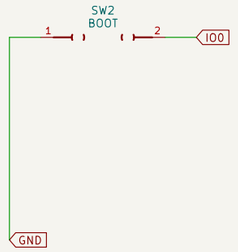


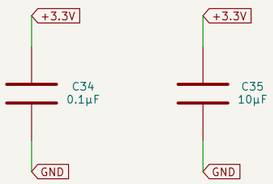




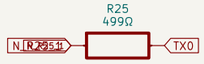




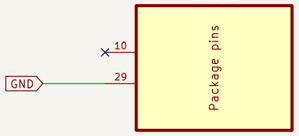












USB−C: USB 2.0 device / 5 V sink. R102 and R103: independent 5.1 kohm CC pull−downs.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

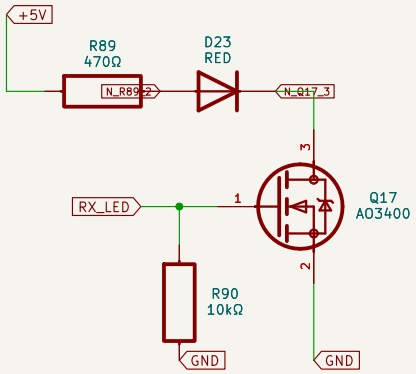
Sheet: /USB + reset / boot/
File: 19_usb.kicad_sch

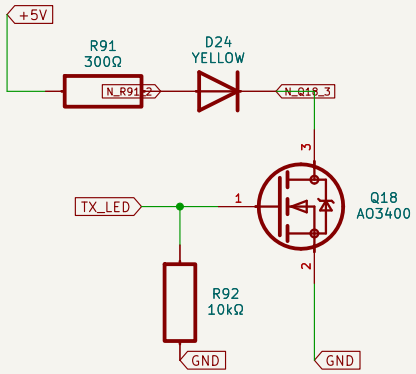
Title: USB + reset / boot

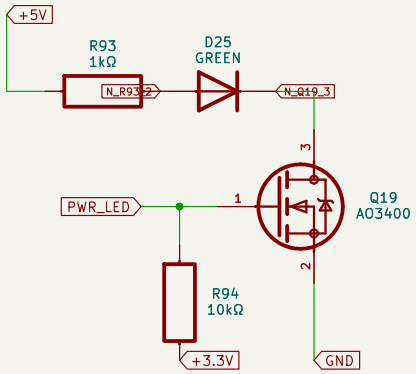
Size: A3	Date: 2026−10−13	**Rev: V240−C/OBD**
KiCad E.D.A. 10.0.7		Id: 20/23

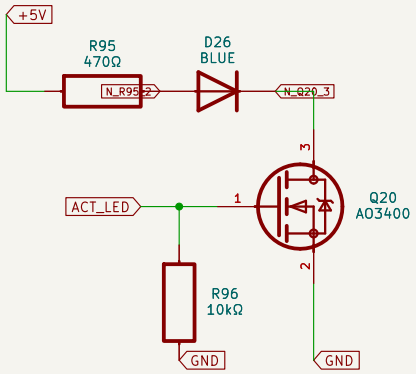
Status LED drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.









Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

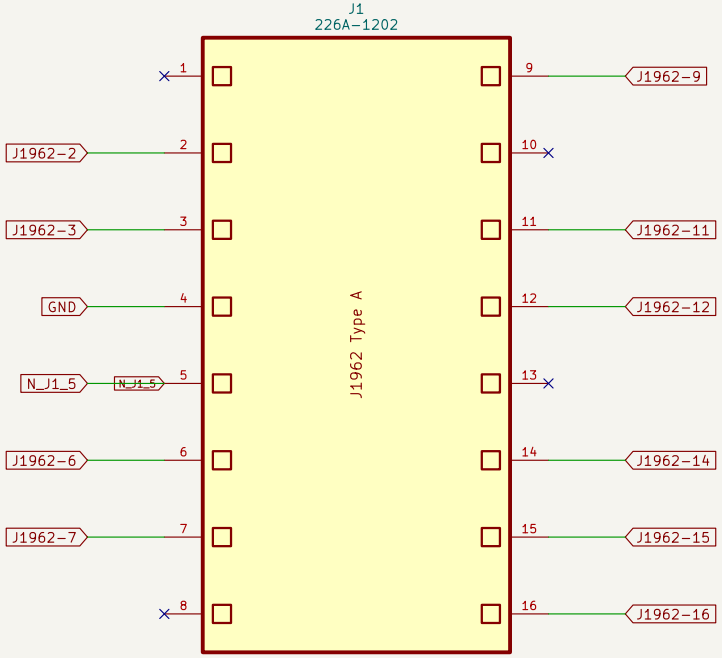
Sheet: /Status LED drivers/
File: 20_leds.kicad_sch

Title: Status LED drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 21/23

OBD connector

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



ATTEND 226A-1202 / J1962 Type A: pins 1-8 near the PCB edge; pins 9-16 inboard.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

Sheet: /OBD connector/
File: 21_obd.kicad_sch

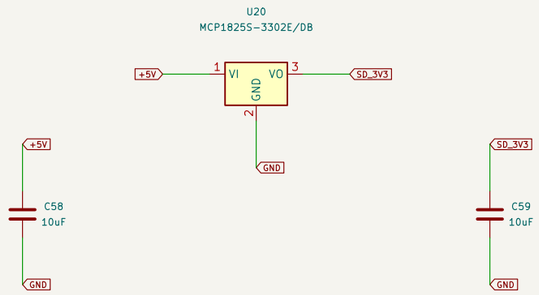
Title: OBD connector

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.7		Id: 22/23

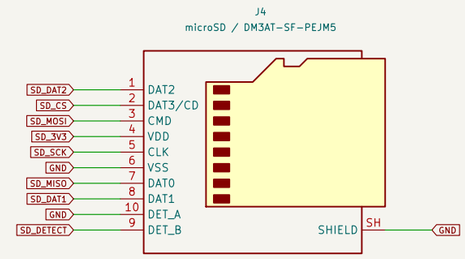
On-board microSD logger (SPI)

Existing ESP32-PICO-V3-02 | removable storage | native scanner functions retained | firmware deferred

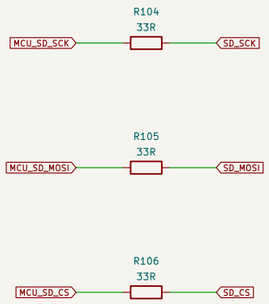
Dedicated 3.3 V SD supply from the existing +5 V rail



J4: Hirose DM3AT-SF-PEJM5 / SPI mode

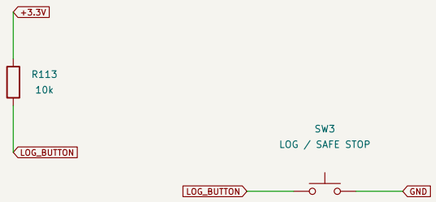


MCU-side series damping

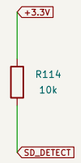


SD CMD/DAT pull-ups / MCU CS strap bias

Active-low record / safe-stop input



Card-detect input



GPIO15 CS is biased high at reset; GPIO12 and GPIO2 remain unused.

GPIO36/37/38 are input-only and have external pull-ups.

GPIO20 is package-specific; use ESP-IDF support for ESP32-PICO-V3.

SD rail transient current and total +5 V budget need hardware validation.

Bulk capacitance is not a safe-power-off hold-up supply.

Firmware deferred: this hardware revision does not yet record data.

Future status indication reuses existing ACT_LED / GPIO13.

Existing sheets use passive visualization pins; native ERC is not independent certification of the scanner circuit.

ChryslerScanner / local hardware revision

Sheet: /Integrated SD logger/
File: 22_sd-logger.kicad_sch

Title: Integrated microSD logging hardware

Size: A3	Date: 2026-10-20	**Rev: V240-SD-A**
KiCad E.D.A. 10.0.7		Id: 23/23